

**FIRST SEMESTER B.A./B.Sc. DEGREE EXAMINATION
NOVEMBER 2020**

(CUCBCSS)

Mathematics

MAT 1B 01—FOUNDATIONS OF MATHEMATICS

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes

Total No. of Questions : 20

Maximum : 20 Marks

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MAT 1B 01—FOUNDATIONS OF MATHEMATICS

(Multiple Choice Questions for SDE Candidates)

1. If A and B are two disjoint sets, then $A \oplus B =$ _____.
 - (A) A.
 - (B) $A \cap B$.
 - (C) $A \cup B$.
 - (D) B.
2. If $A = \{1, 3, 5\}$ and $B = \{1, 2, 3\}$, then $A \oplus B =$ _____.
 - (A) $\{5\}$.
 - (B) $\{2\}$.
 - (C) $\{2, 5\}$.
 - (D) $\{1, 2, 3, 5\}$.
3. Who is considered to be the father of set theory ?
 - (A) Bertand Russel.
 - (B) George Cantor.
 - (C) Srinivasa Ramanujan.
 - (D) George Boole.
4. A and B are subsets of a universal set having 12 elements. If A has 7 elements, B has 9 elements and $A \cap B$ has 9 elements, then what is the number of elements in $A \cup B$?
 - (A) 11.
 - (B) 16.
 - (C) 22.
 - (D) 10.
5. For any two sets A and B, $A - B$ defined by :
 - (A) $\{x : x \in A \text{ and } x \in B\}$.
 - (B) $\{x : x \in A \text{ and } x \notin B\}$.
 - (C) $\{x : x \notin A \text{ and } x \in B\}$.
 - (D) $\{x : x \in A \text{ or } x \in B\}$.
6. If $|A| = 24$, $|B| = 69$ and $|A \cup B| = 81$, then $|A \cap B| =$ _____.
 - (A) 12.
 - (B) 10.
 - (C) 14.
 - (D) 15.
7. If $A = \{1, 2, 3, 4\}$, then the number of non-empty subsets of A is :
 - (A) 16.
 - (B) 15.
 - (C) 32.
 - (D) 3.

8. For any three sets A, B, C, $(A \cup (B \cap C)) = \text{_____}$.
- (A) $(A \cup B) \cap (A \cup C)$. (B) $(A \cup B) \cup (A \cup C)$.
- (C) $(A \cap B) \cap (A \cap C)$. (D) None of these.
9. If a set A has 3 elements and B has 6 elements, then the minimum number of elements in $A \cup B$ is :
- (A) 6. (B) 3.
- (C) 9. (D) None of these.
10. For any three sets A, B, C, $A \times (B - C) = \text{_____}$.
- (A) $(A \times B) \cup (A \times C)$. (B) $(A \times B) \cap (A \times C)$.
- (C) $(A \times B) - (A \times C)$. (D) $(A \times C) - (A \times B)$.
11. For any two sets A and B, a relation from A to B is a subset of _____.
- (A) A. (B) B.
- (C) $A \times B$. (D) $B \times A$.
12. If R is a relation from a non-empty set A to a non-empty set B, then :
- (A) $R = A \cap B$. (B) $R = A \cup B$.
- (C) $R = A \times B$. (D) $R \subseteq A \times B$.
13. If A is a finite set containing 'n' distinct elements, then the number of relations on A is :
- (A) 2^n . (B) n^2 .
- (C) 2^{n^2} . (D) $2n$.
14. Let $R = \{(1, 3), (4, 2), (2, 4), (2, 3), (3, 1)\}$ be a relation on $A = \{1, 2, 3, 4\}$, then R is :
- (A) Not symmetric. (B) Transitive.
- (C) A function. (D) Reflexive.
15. If $A = \{a, b\}$ and $B = \{1, 2, 3\}$, then the number of functions from A to B is :
- (A) 2^3 . (B) 3^2 .
- (C) 2×3 . (D) $2 + 3$.

16. If $f(x) = 2x + 3$ and $g(x) = x^2 + 7$, then the values of 'x' for which $g(f(x)) = 8$ are :
- (A) 1, 2. (B) -1, 2.
(C) -1, -2. (D) 1, -2.
17. Let A be a set containing 'n' distinct elements. How many bijections from A to A can be defined ?
- (A) n^2 . (B) $n!$.
(C) n . (D) $2n$.
18. If $f(x) = \frac{1}{\sqrt{2x-4}}$, then its domain is :
- (A) $\mathbb{R} - \{2\}$. (B) $\mathbb{R}$.
(C) $(2, \infty)$. (D) $[2, \infty)$.
19. Which of the following is a Polynomial function ?
- (A) $\frac{x^2 - 1}{x}, x \neq 0$. (B) $x^3 + 3x^2 - 4x + \sqrt{2}x^{-2}, x \neq 0$.
(C) $\frac{3x^3 + 7x - 1}{3}$. (D) $2x^2 + \sqrt{x} + 1$.
20. If for a function $f(x), f(x+y) = f(x) + f(y)$ for all real number 'x' and 'y'. then $f(0) = \text{————}$.
- (A) 1. (B) -1.
(C) 2. (D) 0.

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Mathematics

MAT 1B 01—FOUNDATIONS OF MATHEMATICS

Time : Three Hours

Maximum : 80 Marks

Section A

*Answer all the **twelve** questions.*

Each question carries 1 mark.

1. Fill in the blanks : When A and B are any two sets, $A \oplus B =$ _____.
2. Fill in the blanks : If A and B are any two finite sets, $n(A \setminus B) = n(A) -$ _____.
3. Define a transitive relation on a set.
4. The diagonal elements of the matrix representing a reflexive relation will be _____.
5. Find the domain of the function $f(x) = \frac{1}{\sqrt{1-x^2}}$. _____.
6. Number of one-one functions from a set A of m elements to a set B of n elements when $n > m$ is _____.
7. Find $\lim_{n \rightarrow \infty} \frac{\sin n}{n}$.
8. Give an example of a denumerable set.
9. Define a bijection from the set of even integers to the set of odd integers.
10. Solve for x and y if $(x + 2y, x - 2y) = (8, 2)$.
11. How many rows appear in a truth table for the compound proposition : $p \rightarrow \neg p$.

Turn over

12. Find the complement of the set of all solutions of the quadratic equation $x^2 - 2x + 1 = 0$ with respect to the set of real numbers.

(12 × 1 = 12 marks)

Section B

Answer any **nine** out of twelve questions.

Each question carries 2 marks.

13. Construct a relation R on $A = \{1, 2, 3\}$ such that R is symmetric and anti-symmetric but not reflexive.
14. Find the matrix of the relation R from $A = \{1, 2, 3, 4\}$ to $B = \{x, y, z\}$ given by $R = \{(1, y), (1, z), (3, y), (4, x), (4, z)\}$.
15. Evaluate the limit of $f(x) = \frac{|x| + x}{x - |x|}$ as x tends to -1 .
16. Find the function obtained by shifting the graph of $f(x) = |x - 1|$ right by 2 units.
17. Test whether the function $\mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^2$ is injective or not.
18. Find all real values of x at which $f(x) = \tan x$ is discontinuous.
19. Show that the relation “is a divisor of” is not a partial order on the set of integers.
20. Find the number of functions which are one-one from $A = \{1, 2, 3\}$ to the set $B = \{a, b, c, d\}$.
21. Find f^{-1} when $f(x) = 2x - 3$.
22. Write down the power set of $A = \{x, y, z\}$.
23. Show that limit of a constant function is that constant through out the domain of that function using the formal definition of the limit.
24. Use a quantified statement to express the verbal statement “Every student in your class has taken a course in calculus.”

(9 × 2 = 18 marks)

Section C

Answer any **six** out of nine questions.

Each question carries 5 marks.

25. Find $g \circ f$ and $f \circ g$, if $f(x) = x^2$ and $g(x) = x + 3$.
26. Show that $\neg \forall x (P(x) \rightarrow Q(x))$ and $\exists x (P(x) \wedge \neg Q(x))$ are logically equivalent.
27. Show that $\mathbb{Q}$, the set of rational numbers is countable.
28. Discuss the continuity of the function $\sin\left(\frac{1}{x}\right)$ at the origin.
29. If $\mathcal{A} = \{\{1, 2, 3\}, \{2, 3, 4\}, \{2, 3, 5\}\}$, find $\bigcup_{A \in \mathcal{A}} A$ and $\bigcap_{A \in \mathcal{A}} A$.
30. Draw the graph of the function obtained by shifting the graph of $f(x) = x^2 - 1$ up by one unit.
31. Let $f: A \rightarrow B$ and $g: B \rightarrow C$ be functions. If $g \circ f$ is onto, show that g is onto.
32. Show that $\neg(p \vee q)$ and $\neg p \wedge \neg q$ are logically equivalent.

(6 × 5 = 30 marks)

Section D

Answer any **two** out of three questions.

Each question carries 10 marks.

34. (a) Find $\lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{x - 4}$.
- (b) Prove that subset of a countable set is countable.
35. (a) Find the continuous extension of the function $h(x) = \frac{x^2 - 4}{x - 2}, x \neq 2$.
- (b) Find $\lim_{x \rightarrow 0^+} \frac{x - \sin mx}{mx}$ when $m \neq 0$.

Turn over

36. (a) What is the truth value of $\forall x (x^2 \geq x)$ if the domain consists of all real numbers? What is the truth value of this statement if the domain consists of all integers?
- (b) Test whether the function

$$f(x) = \begin{cases} \frac{e^{1/x} - 1}{e^{1/x} + 1}, & \text{if } x \neq 0 \\ 0, & \text{otherwise} \end{cases}$$

is continuous or not.

(2 × 10 = 20 marks)